

REPORT SUBMISSION

Design, Modeling, and Simulation of a Bottle Water Rocket

Propulsion, Trajectory, and Static Stability Analysis

Simulation Evidence (MATLAB)



Submitted by

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Declaration

I hereby declare that this report is my own work. All sources of information used in this report have been acknowledged, and any material taken from other sources has been properly cited.

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Abstract

This report presents a simulation-based design and analysis of a 2 L bottle water rocket, focusing on the mandatory deliverables: (i) propulsion/thrust characterization, (ii) flight trajectory simulation, and (iii) static stability verification. A physics-based model is implemented in MATLAB including adiabatic pressure decay, nozzle mass flow, thrust generation, quadratic drag, and point-mass flight dynamics.

For the baseline configuration (Case A), the simulation predicts a peak thrust of 250.07 N, a burn time of 0.13 s, and a total impulse of 16.86 N s. The trajectory reaches an apogee of 22.65 m at 1.80 s, with a total flight time of 4.31 s. Static stability is assessed using a CG–CP static margin; the full-bottle margin is 1.33 cal whereas the empty-bottle margin reduces to 0.08 cal, highlighting the need for fin sizing and/or ballast to maintain stability throughout the flight.

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Chapter 1

Introduction

Water rockets offer a compact and low-cost platform to study the core physics of rocket flight while still exhibiting realistic multi-domain interactions. A typical water rocket uses compressed air as the energy source and water as a reaction mass, producing a short-duration, high-thrust boost followed by a ballistic coast phase. Although the hardware is simple, the flight dynamics are nontrivial because the system combines (i) highly transient thrust generation, (ii) strong mass variation during the expulsion phase, and (iii) aerodynamic forces that scale with the square of velocity. As a result, even small changes in initial pressure, nozzle diameter, water fill fraction, or drag assumptions can produce measurable differences in burnout speed, apogee, and time-of-flight.

The primary objective of this project is to design, model, and evaluate a 2 L bottle water rocket using *software simulation only* and to present simulation evidence that satisfies the required deliverables. The deliverables targeted in this report are:

- (D1) **Trajectory prediction:** compute the time histories of position, velocity, and altitude, and report performance metrics such as apogee and time-of-flight.
- (D2) **Propulsion/thrust analysis:** derive and simulate the bottle pressure evolution, mass flow rate, thrust profile, burnout time, and total impulse.
- (D3) **Stability assessment:** evaluate the static stability margin by comparing the center of gravity (CG) and center of pressure (CP) locations and reporting the resulting static margin across relevant mass states.

A simulation-first approach is selected for three technical reasons. First, the objective requires *detailed thrust and trajectory analysis*, and the water rocket’s propulsion process occurs on a very short time scale (order of 0.1 s in the present configuration). Accurately measuring thrust and mass flow experimentally over such a short duration typically requires dedicated instrumentation (e.g., a calibrated load cell test stand with high sampling rate) and careful repeatability control, which is beyond the scope of this submission. Second, stability verification requires CG–CP analysis at multiple states (e.g., ‘full bottle’ and ‘empty bottle’) and is most transparently demonstrated through a parametric model that explicitly accounts for the mass distribution and fin geometry. Third, simulation enables systematic *verification and sensitivity analysis*: by sweeping parameters (pressure, water fraction, nozzle diameter, drag coefficient), the model can establish cause–effect relationships and quantify how uncertainties propagate to flight performance.

The methodology adopted in this report follows a modular modeling workflow. A propulsion model is first developed to compute thrust as a function of time from the internal bottle pressure and nozzle flow, while simultaneously tracking the changing rocket mass. This thrust model is then coupled to a trajectory model formulated as a set of ordinary differential equations (ODEs) in two/three dimensions, including gravity and aerodynamic drag. In parallel, a static stability

model is used to compute CG and CP and to report a static margin (in calibers) for the relevant operating conditions. The outputs of these models are presented as evidence in the form of (i) time-domain plots for pressure, thrust, mass flow rate, and mass, (ii) trajectory plots and performance summaries, and (iii) stability tables reporting CG, CP, and static margin. These artifacts constitute the verification evidence required by the deliverables.

While a physical prototype is not constructed in this submission, the report is structured so that experimental validation can be incorporated later with minimal changes. Specifically, the simulation artifacts define measurable quantities (burn time, peak thrust, total impulse, apogee) that can be compared directly against future test data, enabling a clear pathway from modeling to validation.

Chapter 2

Requirements and Deliverables

This report is structured to directly satisfy the mandatory project deliverables using a simulation-first workflow for a 2L bottle water rocket. Each deliverable is addressed through clearly presented simulation evidence (figures and professionally formatted tables) generated from the implemented model. Where detailed derivations and step-by-step calculations are required, they are provided in the relevant modeling chapters (System Modeling, Propulsion Modeling, Stability Analysis, and Trajectory Simulation), rather than repeated here.

1. **Software simulation (required):** A complete simulation that produces time histories and plots for bottle pressure, thrust, mass flow rate, rocket mass, and the flight trajectory.
2. **Detailed trajectory analysis:** Altitude vs. time, speed/velocity vs. time, and the flight path, with key events identified (boost/burnout, apogee, and landing).
3. **Thrust/propulsion analysis:** A physically motivated propulsion model producing $T(t)$, $P(t)$, and $\dot{m}(t)$, along with key propulsion metrics such as burn time and total impulse.
4. **Stability analysis:** A static stability assessment using the center of gravity (CG), center of pressure (CP), and the resulting static margin, evaluated for relevant mass states (e.g., full vs. empty).

All deliverables are demonstrated for the baseline configuration (**Case A**). The repository is parameterized so that additional cases can be evaluated by updating the configuration and re-running the simulation to regenerate the corresponding plots and result tables.

Chapter 3

System Modeling

This chapter defines the physical configuration of the 2L bottle water rocket and establishes the modeling assumptions used consistently across propulsion, stability, and trajectory simulation. The intent is to specify the geometry, coordinate conventions, and force models in a traceable manner so that the simulation outputs can be interpreted and reproduced.

3.1 Geometry and parameters

The rocket is based on a standard 2L PET bottle and is modeled as a slender axisymmetric body with an equivalent length of 0.320 m and outer diameter of 0.105 m. A nozzle is assumed at the aft end, and a fin set provides aerodynamic stabilization. The baseline operating condition (Case A) uses an initial water fill fraction of 0.35 and an initial *absolute* bottle pressure of 5.01 bar. Additional parameters such as nozzle diameter, dry mass, fin geometry, and reference area are taken from the project configuration and are reported in the corresponding parameter table in the Simulation Results chapter for transparency and repeatability.

3.2 Coordinate frame and states

A right-handed inertial frame is adopted. The z -axis represents altitude (positive upward) and the x -axis represents downrange distance (positive in the launch direction). The rocket's translational motion is modeled using a point-mass approximation, where the state consists of position and velocity components in the inertial frame. The thrust vector is aligned with the prescribed launch direction during the initial boost, consistent with a guided launch/rail assumption. Attitude dynamics are not explicitly integrated in the baseline model; instead, stability is handled through a static margin check to justify the alignment assumption during nominal flight.

3.3 External forces

The primary external forces considered in the simulation are thrust, gravity, and aerodynamic drag. Gravity is modeled as a constant acceleration g acting in the negative z -direction. Thrust is generated by the propulsion model and is applied along the launch direction during the powered phase. Aerodynamic drag is applied opposite to the instantaneous velocity vector and is modeled using a quadratic drag law.

3.4 Aerodynamic drag model

Aerodynamic drag is represented by

$$D = \frac{1}{2}\rho_a C_D A_{\text{ref}} v^2, \quad (3.1)$$

where ρ_a is the ambient air density, C_D is an effective drag coefficient, A_{ref} is the reference area, and v is the rocket speed. The coefficient C_D is treated as a modeling parameter reflecting the combined effects of body shape, fins, and surface finish; it is selected consistently with typical small-scale rocket values and can be adjusted to evaluate sensitivity. The reference area is taken as the frontal area based on the bottle diameter.

3.5 Time-varying mass

A key feature of water rocket dynamics is the rapid change in mass during the powered phase. The vehicle mass decreases due to the expulsion of water and, depending on the modeled regime, additional air discharge during the blowdown phase. This time-varying mass $m(t)$ is provided by the propulsion model and is coupled directly into the flight dynamics. In the baseline simulation, the mass evolution is tracked continuously to ensure that both thrust-to-weight ratio and acceleration are computed consistently throughout the boost and coast phases.

Chapter 4

Propulsion Modeling

The water rocket propulsion process is modeled as a two-regime system: (i) *water expulsion* driven by compressed air inside the bottle, followed by (ii) *air blowdown* after the water is depleted. The objective of this chapter is to present the governing equations used to compute the time-varying bottle pressure, mass flow rate, thrust, and vehicle mass. An explicit numerical example for the baseline configuration (Case A) is included to demonstrate how the simulation outputs are produced and to support the reported propulsion evidence.

4.1 Model assumptions

The following assumptions are used in the propulsion model:

- The bottle contains compressed air and water; the air expansion during water expulsion is approximated as adiabatic with ratio of specific heats $\gamma \approx 1.4$.
- Ambient pressure is P_a and ambient air density is constant for the propulsion time scale.
- Nozzle losses are represented by a discharge coefficient C_d , applied to the ideal exit velocity.
- Thrust is dominated by momentum thrust in the simplified water phase; the pressure-thrust term is neglected due to near-ambient nozzle exit pressure.

4.2 Phase I: Water expulsion

During water expulsion, the internal air expands as water leaves the bottle. Let V_b be the bottle volume, $V_w(t)$ the water volume, and $V_{air}(t) = V_b - V_w(t)$. Under adiabatic expansion,

$$P_b(t) V_{air}(t)^\gamma = P_0 V_{air,0}^\gamma, \quad \gamma \approx 1.4, \quad (4.1)$$

where $P_b(t)$ is the bottle pressure, P_0 is the initial absolute pressure, and $V_{air,0}$ is the initial trapped air volume.

The effective nozzle exit velocity is modeled using a Bernoulli-type pressure drop with discharge coefficient C_d :

$$v_e(t) = C_d \sqrt{\frac{2(P_b(t) - P_a)}{\rho_w}}, \quad (4.2)$$

where ρ_w is the water density and A_n is the nozzle area. The water mass flow rate is

$$\dot{m}_w(t) = \rho_w A_n v_e(t). \quad (4.3)$$

The water-phase thrust is then modeled as momentum thrust:

$$T(t) \approx \dot{m}_w(t) v_e(t). \quad (4.4)$$

4.3 Phase II: Air blowdown

After water depletion, the remaining compressed air discharges through the nozzle. Compressible discharge is modeled with choked/un-choked flow depending on the bottle-to-ambient pressure ratio. Choking occurs when

$$\frac{P_a}{P_b} \leq \left(\frac{2}{\gamma + 1} \right)^{\frac{\gamma}{\gamma - 1}}. \quad (4.5)$$

For choked flow, the air mass flow rate can be written as

$$\dot{m}_a(t) = C_d A_n P_b(t) \sqrt{\frac{\gamma}{R T_b(t)}} \left(\frac{2}{\gamma + 1} \right)^{\frac{\gamma + 1}{2(\gamma - 1)}}, \quad (4.6)$$

where R is the specific gas constant for air and $T_b(t)$ is the bottle air temperature (handled consistently with the chosen thermodynamic assumptions). For un-choked flow, the standard isentropic nozzle relation is used with the corresponding pressure ratio term. The thrust in the air phase is computed from momentum thrust, optionally including the pressure term when modeled:

$$T(t) \approx \dot{m}_a(t) v_e(t) + (P_e(t) - P_a) A_n. \quad (4.7)$$

In the present implementation, the air phase is included to capture the short-duration tail of thrust after water depletion.

4.4 Mass evolution

The rocket mass varies with expelled water and air:

$$m(t) = m_0 - \int_0^t \dot{m}(\tau) d\tau, \quad (4.8)$$

where $\dot{m}(t) = \dot{m}_w(t)$ in Phase I and $\dot{m}(t) = \dot{m}_a(t)$ in Phase II. This time-varying mass is passed directly to the trajectory solver so that acceleration is computed consistently during boost and coast.

4.5 Case A numerical example (background calculation)

This subsection demonstrates the *actual* numerical calculations behind the generated propulsion curves for the baseline configuration (Case A). The values below correspond to the simulation artifacts reported in the results.

Initial conditions (Case A)

- Bottle volume: $V_b = 2 \text{ L} = 2.0 \times 10^{-3} \text{ m}^3$
- Water fill fraction: $f_w = 0.35 \Rightarrow V_{w,0} = 0.35 V_b = 0.7 \text{ L}$
- Initial absolute pressure: $P_0 = 5.01325 \times 10^5 \text{ Pa}$
- Ambient pressure: $P_a = 1.01325 \times 10^5 \text{ Pa}$
- Water density: $\rho_w = 1000 \text{ kg/m}^3$
- Nozzle discharge coefficient: $C_d = 0.95$
- Nozzle diameter (inferred/used): $d_n = 21 \text{ mm} \Rightarrow A_n = \frac{\pi d_n^2}{4} = 3.4636 \times 10^{-4} \text{ m}^2$

Step 1: Initial exit velocity

At $t = 0$, the pressure differential is $\Delta P = P_0 - P_a = 4.0 \times 10^5$ Pa. Thus,

$$v_e(0) = C_d \sqrt{\frac{2\Delta P}{\rho_w}} = 0.95 \sqrt{\frac{2(4.0 \times 10^5)}{1000}} \approx 26.87 \text{ m/s}.$$

Step 2: Initial mass flow rate

$$\dot{m}_w(0) = \rho_w A_n v_e(0) = 1000 \times 3.4636 \times 10^{-4} \times 26.87 \approx 9.3067 \text{ kg/s}.$$

Step 3: Initial thrust

$$T(0) \approx \dot{m}_w(0) v_e(0) = 9.3067 \times 26.87 \approx 250.07 \text{ N}.$$

This matches the reported peak thrust in the propulsion time history for Case A.

Step 4: Water depletion and burn duration

The simulation indicates that water expulsion completes at approximately $t \approx 0.094$ s, after which the model transitions to air blowdown. A practical burn time is defined as the duration over which thrust remains above a small threshold (numerical robustness). For Case A, the burn ends at approximately $t_b \approx 0.13$ s, consistent with the thrust curve tail in the air phase.

Step 5: Total impulse (numerical integration)

Total impulse is computed by numerically integrating the simulated thrust curve:

$$I_{\text{tot}} = \int_0^{t_b} T(t) dt \approx \sum_{k=0}^{N-1} \frac{T_k + T_{k+1}}{2} \Delta t.$$

For Case A, this yields

$$I_{\text{tot}} \approx 16.86 \text{ N} \cdot \text{s},$$

which is the value reported in the simulation summary and used as propulsion evidence in Chapter 7.

4.6 Computed propulsion outputs

The propulsion simulation outputs the time histories $P_b(t)$, $\dot{m}(t)$, $m(t)$, and $T(t)$. From these, the key propulsion metrics reported in the results chapter are the peak thrust, burn time, and total impulse. The corresponding plots (pressure, thrust, mass flow rate, and mass) are provided as simulation evidence for Case A.

Chapter 5

Static Stability Analysis

Static stability is evaluated using the longitudinal *static margin* (SM), which indicates whether the vehicle tends to restore itself to the direction of flight after a small disturbance. In this report, SM is reported in *calibers*, i.e., normalized by the body diameter.

5.1 Static margin definition

The static margin is defined as

$$SM = \frac{x_{CP} - x_{CG}}{D}, \quad (5.1)$$

where x_{CP} is the center-of-pressure location, x_{CG} is the center-of-gravity location, and D is the body diameter. A positive SM ($x_{CP} > x_{CG}$) indicates static stability, while values close to zero represent near-neutral stability.

5.2 CG and CP estimation

The center of gravity (CG) is computed using a mass-weighted centroid model of the major components, including the bottle structure, water payload, and the nose/fin assemblies. The center of pressure (CP) is estimated using a Barrowman-style slender-body approximation (nose contribution plus fin contribution), which is appropriate for a first-order static margin check in the absence of wind-tunnel or CFD data.

5.3 Results

Table 5.1 summarizes the computed stability metrics for the two critical operating states: *full bottle* (maximum water mass at launch) and *empty bottle* (after propellant depletion). The CP is a geometry-driven quantity and is therefore identical across the two mass states, whereas the CG shifts significantly as the water mass is expelled.

Table 5.1: Static stability metrics for Case A (reported in meters and calibers).

Metric	Full bottle	Empty bottle
x_{CG} [m]	0.069	0.200
x_{CP} [m]	0.208	0.208
Static margin [cal]	1.330	0.080

The full-bottle configuration is comfortably stable with a static margin above one caliber, which supports the assumption of nominal alignment during the boost phase. After depletion, the

static margin reduces to a near-neutral value, indicating that practical designs should maintain a positive margin throughout the full flight envelope. This can be achieved by increasing fin effectiveness (e.g., larger fin area or aft fin placement) and/or adding a small nose ballast to prevent the CG from moving too far aft during the coast phase.

Chapter 6

Trajectory Simulation

This chapter describes the flight-dynamics model used to simulate the 2L bottle water rocket trajectory. A point-mass formulation is adopted to predict the translational motion under the combined influence of time-varying thrust, aerodynamic drag, and gravity. The propulsion model (Chapter 4) provides the thrust history and time-varying mass, which are coupled directly into the trajectory integration.

6.1 Equations of motion

A 3D point-mass model is used with state vector $\mathbf{x} = [x, y, z, v_x, v_y, v_z, m]^T$, where $[x, y, z]^T$ is position, $[v_x, v_y, v_z]^T$ is velocity, and $m(t)$ is the time-varying mass. The translational dynamics are expressed as

$$\dot{\mathbf{v}}(t) = \frac{1}{m(t)} (\mathbf{T}(t) - \mathbf{D}(t)) + \mathbf{g}, \quad \dot{\mathbf{r}}(t) = \mathbf{v}(t), \quad \dot{m}(t) = -\dot{m}(t), \quad (6.1)$$

where $\mathbf{g} = [0, 0, -g]^T$. Aerodynamic drag is modeled as a quadratic force opposite to the direction of motion, and the thrust vector is applied along the prescribed launch direction during the powered phase. This formulation captures the dominant effects governing apogee and time-of-flight while keeping the model sufficiently transparent for verification and reproducible simulation evidence.

6.2 Numerical integration and event handling

The coupled ODE system is integrated using a variable-step Runge–Kutta method (MATLAB `ode45`). To ensure a physically meaningful end condition, an event function is used to terminate the simulation when the rocket returns to ground level (i.e., when $z(t) = 0$ on descent). Key flight events are handled consistently within the simulation workflow:

- **Launch rail segment:** Initial motion is constrained along the launch direction until the rocket exits the rail. This approximates guided launch behavior and prevents unrealistic lateral divergence during the very early stage of flight.
- **Powered-to-coast transition:** The thrust input follows the propulsion model output. After burnout, thrust is set to zero and the rocket continues under gravity and drag only.
- **Landing termination:** The integration stops at ground contact on descent to provide a consistent time-of-flight measure and avoid non-physical negative altitudes in plots and tables.

6.3 Simulation outputs and extracted metrics

The trajectory simulation produces time histories and curves that are directly used as deliverable evidence:

- Altitude $z(t)$ and speed $\|\mathbf{v}(t)\|$ versus time.
- Downrange distance and full 3D flight path $(x(t), y(t), z(t))$.
- Automatically extracted key events including burnout, apogee, and landing, used to populate report-ready summary tables.

All plots and summary tables presented in the Simulation Results chapter are generated directly from these simulation outputs, ensuring that the reported performance metrics are traceable to the implemented model.

Chapter 7

Simulation Results

This chapter presents the simulation evidence for the baseline configuration (**Case A**) and reports the outputs required to satisfy the project deliverables. The results are organized into: (i) key performance metrics reported in a concise table, (ii) propulsion time histories demonstrating the transient pressure–thrust–mass flow behavior, and (iii) trajectory time histories demonstrating altitude and speed evolution through boost, coast, and descent. All figures and tables in this chapter are generated directly from the simulation outputs and formatted for report-ready presentation.

7.1 Key performance metrics

Table 7.1 summarizes the main performance indicators obtained from the simulation. These metrics provide a compact view of propulsion strength (peak thrust, burn time, total impulse), flight outcome (apogee, time-of-flight), and operating conditions (initial and end pressures). In addition, the table provides a traceable checklist that the mandatory deliverables are met: propulsion evidence ($T_{\max}$, t_b , I , $\dot{m}_{\max}$, P_0 , P_{end}) and trajectory evidence ($z_{\max}$, t_a , t_f).

Table 7.1: Key simulated performance metrics for Case A.

Metric	Value
Apogee altitude [m]	22.65
Time to apogee [s]	1.80
Time of flight [s]	4.31
Maximum thrust [N]	250.07
Burn time [s]	0.13
Total impulse [N s]	16.86
Maximum mass flow [kg/s]	9.31
Initial bottle pressure P_0 [bar]	5.01
Final bottle pressure P_{end} [bar]	1.03

7.2 Propulsion time histories

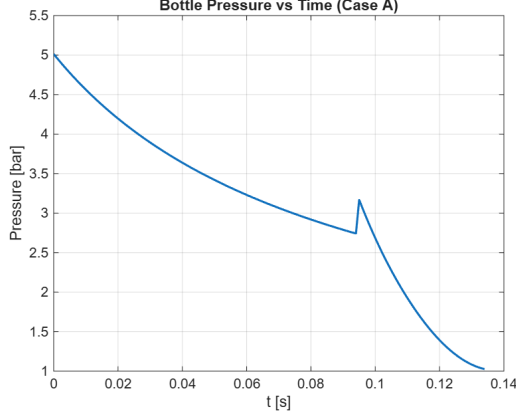
Figure 7.1 presents the propulsion-related time histories: bottle pressure, thrust, mass flow rate, and rocket mass. These plots are the **primary evidence** that the implemented propulsion model is physically consistent and that the simulation captures the key coupled dynamics driving the flight.

Bottle pressure $P_b(t)$ (Fig. 7.1a). This plot shows how the internal bottle pressure decays as the compressed air expands while expelling water. The pressure profile is important because it is the *driving input* to the nozzle flow model: the pressure differential ($P_b(t) - P_a$) determines the exit velocity and therefore the thrust. The observed rapid decay from the initial 5.01 bar toward near-ambient values provides clear evidence of the expected water-rocket behavior (high initial energy followed by a fast discharge process).

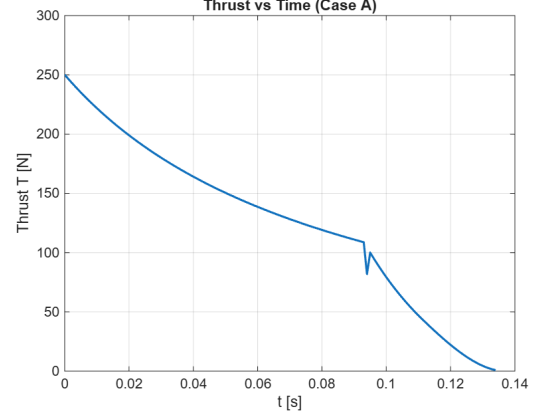
Thrust $T(t)$ (Fig. 7.1b). The thrust plot provides direct evidence for the thrust deliverable. The large initial thrust peak at launch reflects the maximum pressure differential at $t = 0$. The subsequent rapid decay demonstrates that thrust is transient and short-lived, consistent with water expulsion followed by a brief blowdown tail. This curve is also used to define propulsion metrics such as peak thrust and burn time, and it sets the initial acceleration that strongly influences apogee.

Mass flow rate $\dot{m}(t)$ (Fig. 7.1c). The mass flow plot validates that the thrust is produced by physically reasonable expulsion of propellant mass through the nozzle. The high initial $\dot{m}$ indicates rapid water discharge early in the burn, which decreases as the pressure drops. This plot is crucial for evidence because it connects pressure dynamics to thrust via the momentum term and ensures the model is not treating thrust as an arbitrary input.

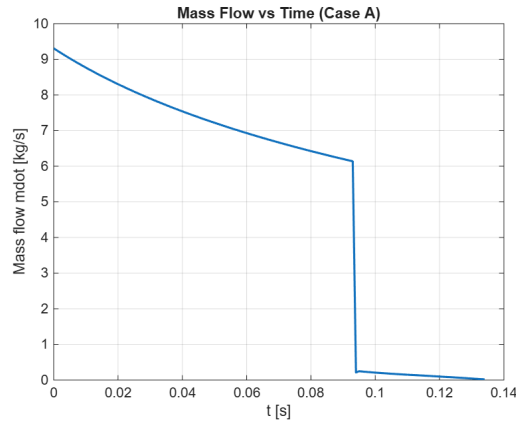
Rocket mass $m(t)$ (Fig. 7.1d). The mass history demonstrates the *time-varying mass* effect, which is a defining feature of water rockets. The rapid drop in mass during the burn is evidence that the simulation correctly couples propellant depletion into the flight dynamics. This is important because acceleration depends on T/m ; a constant-mass assumption would significantly distort the predicted trajectory.



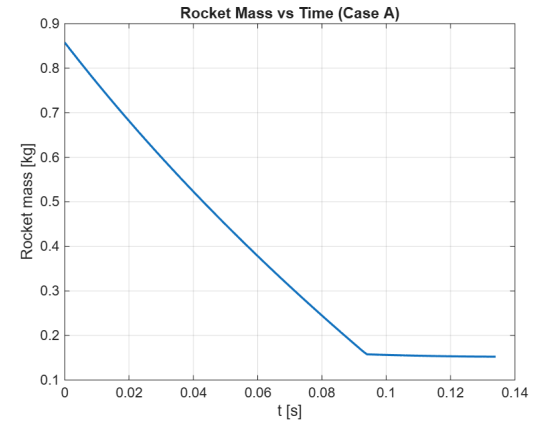
(a) Bottle pressure $P_b(t)$.



(b) Thrust $T(t)$.



(c) Mass flow rate $\dot{m}(t)$.



(d) Rocket mass $m(t)$.

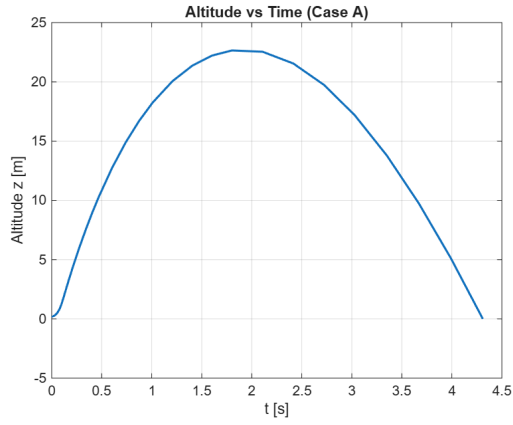
Figure 7.1: Propulsion-related simulation outputs for Case A.

7.3 Flight trajectory and kinematic outputs

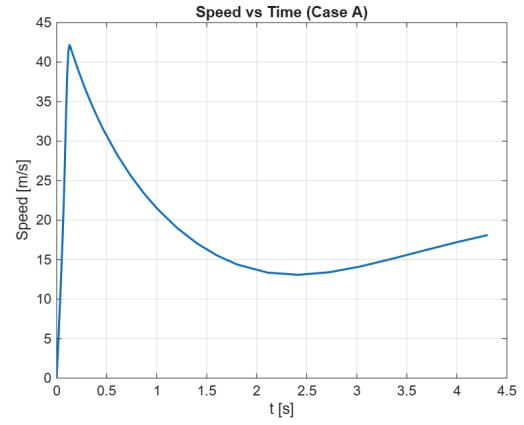
Figure 7.2 provides the required trajectory evidence by showing altitude and speed versus time over the complete flight. These plots demonstrate the transition from powered ascent to ballistic coast and finally to descent, and they support the extracted metrics reported in Table 7.1.

Altitude vs. time (Fig. 7.2a). The altitude curve is the primary trajectory deliverable and provides direct evidence of the full flight profile. The early steep rise corresponds to boost while thrust is present, followed by a reduced slope during coast as vertical velocity decreases under gravity and drag. The peak of this curve defines the apogee altitude and time-to-apogee, while the return to $z = 0$ on descent defines the time-of-flight.

Speed vs. time (Fig. 7.2b). The speed curve explains *why* the altitude evolves as observed. The sharp increase in speed during the initial interval is consistent with the thrust peak and high thrust-to-mass ratio. After burnout, speed decreases due to drag and the conversion of kinetic energy into altitude (and later descent dynamics). This plot provides supporting evidence that propulsion and aerodynamics are coupled correctly and that the model captures the expected acceleration/deceleration phases.



(a) Altitude vs. time.



(b) Speed vs. time.

Figure 7.2: Trajectory simulation outputs for Case A.

Remark: The stability assessment in Chapter 5 complements these trajectory results by verifying that the configuration is statically stable at launch and by highlighting the near-neutral margin after propellant depletion. This provides a complete deliverable-aligned evidence set: propulsion generation, trajectory prediction, and stability verification.

Chapter 8

Discussion

The simulation results demonstrate the characteristic behavior of a 2L bottle water rocket: a short-duration, high-thrust boost phase followed by a coast-dominated trajectory. The propulsion time histories confirm that the thrust is primarily driven by the initial pressure differential across the nozzle. In Case A, the thrust peaks at launch (250.07 N) and decays rapidly as the bottle pressure decreases from 5.01 bar toward ambient. This pressure decay is directly reflected in the mass flow and thrust profiles, producing a sharp early acceleration and setting the initial conditions for the subsequent ballistic motion.

The trajectory outputs are consistent with this propulsion behavior. The speed curve shows a rapid rise during the burn followed by a gradual reduction during coast as aerodynamic drag dissipates kinetic energy and gravity reduces the vertical velocity component. The altitude profile clearly separates the boost phase, coast-to-apogee, and descent-to-landing, and the extracted metrics (apogee, time-to-apogee, and time-of-flight) provide a complete and deliverable-aligned summary of flight performance. Collectively, the pressure, thrust, mass flow, and mass evolution plots provide strong evidence that the implemented model captures the key coupled dynamics required by the project.

From a stability perspective, the analysis highlights an important design consideration for water rockets: the center of gravity shifts substantially as water is expelled. The full-bottle configuration is comfortably stable ($SM \approx 1.33$ calibers), supporting nominal stable flight during the initial boost. However, the empty-bottle static margin reduces to near-neutral ($SM \approx 0.08$ calibers), implying that the vehicle may become sensitive to disturbances late in the flight (e.g., small asymmetries, gusts, or misalignment). For practical robustness, a design target of $SM \gtrsim 1$ caliber throughout the full mass envelope is commonly adopted. This can be achieved through standard corrective actions such as increasing fin area, moving the fins further aft, or introducing a small amount of nose ballast to prevent the CG from shifting too far aft during coast.

The current model also has limitations that define clear directions for future improvement. Aerodynamic drag is represented using a constant effective coefficient C_D , which aggregates multiple effects (body/fin shape, angle of attack, surface roughness) into a single parameter. While sufficient for a baseline trajectory prediction, this simplification can affect apogee accuracy and coast behavior. The thrust formulation also uses a simplified treatment of nozzle exit conditions; incorporating a more detailed pressure-thrust term and loss modeling (or calibrating the discharge coefficient using a thrust-stand measurement) would improve fidelity. Finally, the trajectory model uses a point-mass approximation without explicit attitude dynamics; therefore, pitch and yaw responses to wind, fin restoring moments, and potential weathercocking effects are not captured.

Overall, within the scope of a simulation-only submission, the results provide coherent and traceable evidence for the required deliverables: thrust and propulsion characterization, detailed trajectory prediction, and a static stability assessment identifying both nominal stability at launch and near-neutral margin after depletion.

Chapter 9

Conclusion

This report presented a complete **simulation-only** workflow for the design, modeling, and evaluation of a 2L bottle water rocket. The workflow was structured to satisfy the mandatory project deliverables through traceable simulation evidence, covering (i) propulsion and thrust generation, (ii) trajectory prediction using a coupled point-mass flight model, and (iii) static stability verification using CG–CP and static margin.

For the baseline configuration (Case A), the simulation predicts an apogee of approximately 22.65 m with a short-duration, high-thrust boost (peak thrust 250.07 N) followed by a coast-dominated ascent. The propulsion time histories (pressure, thrust, mass flow, and mass evolution) confirm the expected water-rocket behavior, while the trajectory plots and summary metrics provide the required flight characterization (time-to-apogee and time-of-flight).

The static stability analysis shows that the rocket is comfortably stable at launch ($SM \approx 1.33$ calibers), but the margin decreases substantially after water depletion ($SM \approx 0.08$ calibers), approaching neutral stability. This outcome emphasizes the importance of maintaining a positive static margin across the full mass envelope for practical robustness, and it motivates straightforward design improvements such as increased fin effectiveness and/or small nose ballast.

Overall, the developed repository and report structure provide a reproducible foundation for further work. The same workflow can be extended to parameter sweeps (pressure, water fraction, nozzle size, drag coefficient) and, if desired, to experimental validation by incorporating measured thrust and flight data for calibration and comparison.